

Research Notes on Image segmentation

Image segmentation

>> Section 1

$$P(\lambda \mid f_i) = P(f_i \mid \lambda) P(\lambda) \sum_{\lambda \in \Lambda} P(f_i \mid \lambda) P(\lambda) \\ \{\displaystyle P(\lambda \mid f_i) = \frac{P(f_i \mid \lambda) P(\lambda)}{\sum_{\lambda \in \Lambda} P(f_i \mid \lambda) P(\lambda)}\}$$

>> Section 2

Variational methods The goal of variational methods is to find a segmentation which is optimal with respect to a specific energy functional. The functionals consist of a data fitting term and a regularizing terms. A classical representative is the Potts model defined for an image f by

>> Section 3

Region-growing methods Region-growing methods rely mainly on the assumption that the neighboring pixels within one region have similar values. The common procedure is to compare one pixel with its neighbors. If a similarity criterion is satisfied, the pixel can be set to belong to the same cluster as one or more of its neighbors. The selection of the similarity criterion is significant and the results are influenced by noise in all instances. The method of Statistical Region Merging (SRM) starts by building the graph of pixels using 4-connectedness with edges weighted by the absolute value of the intensity difference. Initially each pixel forms a single pixel region. SRM then sorts those edges in a priority queue and decides whether or not to merge the current regions belonging to the edge pixels using a statistical predicate. One region-growing method is the seeded region growing method. This method takes a set of seeds as input along with the image. The seeds mark each of the objects to be segmented. The regions are iteratively grown by comparison of all unallocated neighboring pixels to the regions. The difference between a pixel's intensity value and the region's mean, δ , is used as a measure of similarity. The pixel with the smallest difference measured in this way is assigned to the respective region. This process continues until all pixels are assigned to a region. Because seeded region growing requires seeds as additional input, the segmentation results are dependent on the choice of seeds, and noise in the image can cause the seeds to be poorly placed. Another region-growing method is the unseeded region growing method. It is a modified algorithm that does not require explicit seeds. It starts with a single region A_1 —the pixel chosen here does not markedly influence the final segmentation. At each iteration it considers the neighboring pixels in the same way as seeded region growing. It differs from seeded region growing in that if the minimum δ

is less than a predefined threshold T then it is added to the respective region A_j . If not, then the pixel is considered different from all current regions A_i and a new region A_{n+1} is created with this pixel. One variant of this technique, proposed by Haralick and Shapiro (1985), is based on pixel intensities. The mean and scatter of the region and the intensity of the candidate pixel are used to compute a test statistic. If the test statistic is sufficiently small, the pixel is added to the region, and the region's mean and scatter are recomputed. Otherwise, the pixel is rejected, and is used to form a new region. A special region-growing method is called λ -connected segmentation (see also λ -connectedness). It is based on pixel intensities and neighborhood-linking paths. A degree of connectivity (connectedness) is calculated based on a path that is formed by pixels. For a certain value of λ , two pixels are called λ -connected if there is a path linking those two pixels and the connectedness of this path is at least λ . λ -connectedness is an equivalence relation. Split-and-merge segmentation is based on a quadtree partition of an image. It is sometimes called quadtree segmentation. This method starts at the root of the tree that represents the whole image. If it is found non-uniform (not homogeneous), then it is split into four child squares (the splitting process), and so on. If, in contrast, four child squares are homogeneous, they are merged as several connected components (the merging process). The node in the tree is a segmented node. This process continues recursively until no further splits or merges are possible. When a special data structure is involved in the implementation of the algorithm of the method, its time complexity can reach $O(n \log n)$, an optimal algorithm of the method.

>> Section 4

Parametric methods Lagrangian techniques are based on parameterizing the contour according to some sampling strategy and then evolving each element according to image and internal terms. Such techniques are fast and efficient, however the original "purely parametric" formulation (due to Kass, Witkin and Terzopoulos in 1987 and known as "snakes"), is generally criticized for its limitations regarding the choice of sampling strategy, the internal geometric properties of the curve, topology changes (curve splitting and merging), addressing problems in higher dimensions, etc.. Nowadays, efficient "discretized" formulations have been developed to address these limitations while maintaining high efficiency. In both cases, energy minimization is generally conducted using a

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steepest-gradient descent, whereby derivatives are computed using, e.g., finite differences.

>> Section 5

Application of Isolated Point Detection in X-ray Image Processing The detection of isolated points has significant applications in various fields, including X-ray image processing. For instance, an original X-ray image of a turbine blade can be examined pixel-by-pixel to detect porosity in the upper-right quadrant of the blade. The result of applying an edge detector's response to this X-ray image can be approximated. This demonstrates the segmentation of isolated points in an image with the aid of single-pixel probes.

>> Section 6

Pick K cluster centers, either randomly or based on some heuristic method, for example K-means++ Assign each pixel in the image to the cluster that minimizes the distance between the pixel and the cluster center Re-compute the cluster centers by averaging all of the pixels in the cluster Repeat steps 2 and 3 until convergence is attained (i.e. no pixels change clusters) In this case, distance is the squared or absolute difference between a pixel and a cluster center. The difference is typically based on pixel color, intensity, texture, and location, or a weighted combination of these factors. K can be selected manually, randomly, or by a heuristic. This algorithm is guaranteed to converge, but it may not return the optimal solution. The quality of the solution depends on the initial set of clusters and the value of K. The Mean Shift algorithm is a technique that is used to partition an image into an unknown apriori number of clusters. This has the advantage of not having to start with an initial guess of such parameter which makes it a better general solution for more diverse cases.